

MARAGHEH, Iran, Islamic Republic ofWMO: **407130**Lat: **37.3477N**Lon: **46.1461E**Elev: **1478**StdP: **84.78**Time Zone: **3.50 (IRN)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.8	-8.2	-16.0	1.1	-8.4	-13.8	1.4	-4.1	9.6	1.7	8.4	3.5	2.0	80	0.541

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	36.8	17.3	35.0	16.8	33.7	16.5	18.8	33.0	18.1	32.1	17.5	31.2	3.4	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
13.4	11.5	26.6	12.5	10.8	25.9	11.8	10.3	25.0	59.9	33.2	57.5	32.0	55.1	31.1	22.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation	n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4) 10.4	9.3	8.2		-13.0	38.9	3.9	1.4	-15.9	39.9	-18.1	40.7	-20.3	41.4	-23.1	42.4	
(5)			DB	-14.0	20.6	3.5	0.8	-16.5	21.2	-18.5	21.6	-20.5	22.0	-23.0	22.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	14.0	0.5	2.8	7.5	12.3	17.9	23.8	27.4	27.3	22.4	15.4	7.5	2.4	(6)
(7)		DBStd	10.09	4.46	4.07	4.09	3.65	2.90	2.62	2.56	2.37	3.10	3.59	3.76	4.47	(7)
(8)		HDD10.0	944	295	202	97	17	0	0	0	0	0	4	92	237	(8)
(9)		HDD18.3	2482	553	434	334	184	44	1	0	0	6	105	326	494	(9)
(10)		CDD10.0	2398	0	1	21	86	244	414	538	537	371	169	16	1	(10)
(11)		CDD18.3	895	0	0	0	3	30	165	280	279	126	12	0	0	(11)
(12)		CDH23.3	9773	0	0	1	31	279	1711	3271	3203	1172	106	0	0	(12)
(13)		CDH26.7	4599	0	0	0	6	44	723	1712	1663	438	13	0	0	(13)
(14)	Wind	WSAvg	3.6	2.5	3.0	3.6	3.7	3.8	4.2	4.8	4.7	3.8	3.3	2.8	2.4	(14)
(15)	Precipitation	PrecAvg	290	29	25	41	58	34	6	2	1	5	22	36	28	(15)
(16)		PrecMax	517	60	87	90	120	94	41	16	5	39	113	92	93	(16)
(17)		PrecMin	180	8	1	10	7	4	0	0	0	0	1	0	2	(17)
(18)		PrecStd	81	14	17	19	30	22	9	4	1	9	29	23	20	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.0	14.9	22.3	26.8	29.9	35.2	39.0	38.2	34.6	28.1	19.8	14.9	(19)
(20)			MCWB	5.0	6.6	9.7	12.5	14.7	17.2	18.4	17.4	15.9	13.3	9.8	6.4	(20)
(21)		2%	DB	10.0	12.5	19.1	23.1	27.9	33.5	37.0	36.7	32.8	26.1	17.2	11.9	(21)
(22)			MCWB	4.4	5.6	8.6	11.2	14.0	16.2	17.8	17.0	15.5	12.6	8.3	5.8	(22)
(23)		5%	DB	8.2	10.5	16.6	20.9	26.2	32.1	35.2	35.1	31.0	24.1	15.1	9.9	(23)
(24)			MCWB	3.8	4.6	7.4	10.5	13.4	15.6	17.1	16.6	14.8	12.0	7.8	5.1	(24)
(25)		10%	DB	6.6	8.8	14.4	19.0	24.8	30.8	33.8	33.8	29.2	22.1	13.2	8.1	(25)
(26)			MCWB	2.9	4.0	6.6	9.5	13.0	15.3	17.0	16.3	14.1	11.2	7.2	4.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.6	7.9	10.6	13.5	16.9	18.6	20.1	19.5	17.3	14.4	11.1	7.8	(27)
(28)			MCDWB	10.2	13.3	19.1	22.4	26.3	33.0	34.4	33.3	30.3	25.4	16.4	12.1	(28)
(29)		2%	WB	5.2	6.5	9.2	12.4	15.4	17.5	19.1	18.6	16.5	13.3	10.0	6.6	(29)
(30)			MCDWB	8.5	10.9	17.2	20.3	25.0	31.3	33.5	32.4	29.4	23.3	14.1	10.7	(30)
(31)		5%	WB	4.1	5.3	8.1	11.4	14.4	16.6	18.5	17.9	15.7	12.6	9.0	5.5	(31)
(32)			MCDWB	7.5	9.2	15.2	18.5	24.1	30.3	32.7	31.9	28.6	21.9	13.3	9.1	(32)
(33)		10%	WB	3.0	4.3	7.1	10.5	13.5	15.9	17.8	17.2	15.0	11.9	8.0	4.4	(33)
(34)			MCDWB	6.2	8.0	13.5	17.0	22.9	29.0	31.6	31.2	27.8	20.4	12.6	7.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.8	8.8	9.8	10.4	11.7	12.8	12.2	12.5	12.6	10.9	9.2	7.7	(35)
(36)			MCDBR	9.3	11.1	12.6	12.8	13.3	13.8	13.2	13.3	13.7	13.3	11.6	10.1	(36)
(37)			MCWBR	5.5	6.2	5.9	5.7	5.6	5.8	5.2	5.4	5.8	5.9	6.2	5.9	(37)
(38)		5% WB	MCDWB	8.3	9.2	11.1	11.0	12.3	12.9	12.3	12.5	12.6	11.9	8.9	8.6	(38)
(39)			MCWBR	5.3	5.5	5.7	5.3	5.6	5.6	5.1	5.3	5.7	5.5	4.9	5.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.310	0.345	0.397	0.466	0.457	0.420	0.454	0.421	0.372	0.386	0.330	0.306	(40)	
(41)		taud	2.319	2.218	2.084	1.927	1.989	2.125	2.039	2.132	2.260	2.163	2.320	2.360	(41)	
(42)		Ebn at Noon	868	876	858	818	831	861	828	849	875	815	836	848	(42)	
(43)		Edh at Noon	94	119	150	186	178	155	168	150	124	124	94	85	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.54	3.47	4.33	5.41	6.74	8.07	7.80	7.15	5.90	4.04	2.72	2.13	(44)	
(45)		RadStd	0.20	0.34	0.37	0.47	0.38	0.38	0.21	0.22	0.31	0.29	0.34	0.28	(45)	